Examiner only

3. *(a)* Give the definition of the *specific heat capacity* of a material. [1]

(b) An insulating flask contains 0.6×10^{-3} m^3 of water at 19.5 °C. A volume 1.6×10^{-3} m^3 of boiling water at 100.0 °C is poured into the flask.

(specific heat capacity of water, c = 4 200 J kg^{-1} °C^{-1}; density of water, ρ = 1 000 kg m^{-3})

(i) Determine the final temperature of the water in the flask. [3]

(ii) Calculate the heat lost by the boiling water. [3]

(iii) Justify the statement that no work is done on/by the boiling water as it cools. No further calculations are needed. [1]

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